

Problem-Solution Fit

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.

1. CUSTOMER SEGMENT(S) [CS] • Smallholder & seasonal farmers • Agricultural researchers • Policymakers & agri-dept officers	6. CUSTOMER LIMITATIONS [CL] • Limited internet access in rural areas • Low digital literacy among older farmers • No budget for premium agri-tech tools	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • Govt advisories: free but generic (con) • Agri consultants: expert but costly (con) • OptiCrop: data-driven & free (pro)
2. PROBLEMS / PAINS + FREQUENCY [PR] • Wrong crop -> yield loss (every season) • Over/under fertilizer use (every cycle) • No real-time soil-climate guidance (daily)	9. PROBLEM ROOT / CAUSE [RC] • No localized data-driven crop advice • Tradition-based decisions over soil science • Soil test data not linked to crop choices	7. BEHAVIOR + ITS INTENSITY [BE] • Asks neighbors for crop tips (high freq) • Plants same crop yearly (high intensity) • Ignores soil test results (medium intensity)
3. TRIGGERS TO ACT [TR] • Start of planting season • Recent crop failure or poor yield • Neighbor succeeds with a different crop	10. YOUR SOLUTION [SL] OptiCrop: AI-powered crop advisor. • Input: N,P,K, temp, humidity, pH, rainfall • Output: Best-fit crop recommendation • Soil-climate suitability check • Fertilizer & resource optimization tips	8. CHANNELS OF BEHAVIOR [CH] ONLINE: Web app, mobile browser, WhatsApp bot OFFLINE: Agri-officer field visits Printed crop-suitability report sheets
4. EMOTIONS BEFORE / AFTER [EM] BEFORE: Anxious, uncertain & overwhelmed about crop decisions & financial risk. AFTER: Confident, informed & optimistic with AI-driven data support.		